

Machine Learning Lab Programs

Lab 1: Implement and demonstrate the FIND – S Algorithm for finding the most specific hypothesis based on a given set of training data samples.

Read the training data from a CSV file

```
import pandas as pd
```

```
# Read the CSV file
```

```
data = pd.read_csv("training_data.csv")
```

```
print("Training Data:")
```

```
print(data)
```

```
# Convert DataFrame to list
```

```
concepts = data.iloc[:, :-1].values.tolist()
```

```
target = data.iloc[:, -1].values.tolist()
```

```
# FIND-S Algorithm
```

```
def find_s(concepts, target):
```

```
    # Initialize hypothesis
```

```
    hypothesis = ['0'] * len(concepts[0])
```

```
print("\nInitial Hypothesis:")

print(hypothesis)

# Process each training example
for i in range(len(concepts)):

    if target[i] == "Yes":

        for j in range(len(hypothesis)):

            if hypothesis[j] == '0':

                hypothesis[j] = concepts[i][j]

            elif hypothesis[j] != concepts[i][j]:

                hypothesis[j] = '?'

        print(f"\nHypothesis after Positive Example {i+1}:")

        print(hypothesis)

return hypothesis
```

```
# Find the final hypothesis
```

```
final_hypothesis = find_s(concepts, target)
```

```
print("\nFinal Hypothesis:")
```

```
print(final_hypothesis)
```

Output:

Training Data:

	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes

Initial Hypothesis:

```
['0', '0', '0', '0', '0', '0']
```

Hypothesis after Positive Example 1:

```
['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
```


Hypothesis after Positive Example 2:

['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Hypothesis after Positive Example 4:

['Sunny', 'Warm', '?', 'Strong', '?', '?']

Final Hypothesis:

['Sunny', 'Warm', '?', 'Strong', '?', '?']